
RESEARCH NOTE

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The Effects of a Body-Focused Physical and Health Education Module on Self-Objectification and Social Physique Anxiety in Irish Girls

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The purpose of this study was to examine the effects of an 8-week body-focused physical and health education module on self-objectification and social physique anxiety (SPA) in a sample of 85 Irish schoolgirls. Classrooms were randomly assigned to receive the experimental module or the standard curriculum. Participants completed pre- and postassessments of the value they placed on objectifying and nonobjectifying physical attributes, along with a measure of SPA. Girls in the experimental condition increased the value they placed on physical health and strength, decreased the value they placed on sex appeal, and showed no change in SPA. Girls in the control condition decreased the value they placed on body weight and physical fitness and experienced a significant increase in SPA. These results suggest that a body-focused module can decrease self-objectification and prevent developmentally linked increases in SPA.

Keywords: girl-friendly physical education, body image, psychological well-being, pedagogy

It is well documented that boys are more physically active than girls. In physical activity surveys conducted in industrialized nations (for a review, see Dishman, Washburn, & Heath, 2004), boys are invariably shown to engage in more weekly minutes of activity than girls of the same age. In order to explain these gender differences, girls themselves have often been identified and targeted for investigation as "the problem," and have even been blamed for their lack of participation. Unfortunately, few researchers or interventionists have considered that the problem of female nonparticipation may lie with the types of physical activity offerings that are provided for girls (Ennis, 1999). Indeed, it has been suggested that "the key challenge facing the education and sport communities is to develop forms of

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physical education and sport that are sensitive and specific to girls' needs, in fact to develop and practice programmes of physical education that are 'girl friendly'" (Kirk, Fitzgerald, Wang, & Biddle, 2000, p. 3).

Girl-friendly physical education programs aim to address the gender imbalance in physical education participation by implementing policies and procedures that encourage girls' participation. Such programs acknowledge that traditional methods of delivering physical education (technocratic teaching styles, an emphasis on competitive sport, and technique development) are not particularly meaningful or enjoyable to girls (Kirk et al., 2000). These programs also recognize that girls' concerns about exposing their bodies and having their bodies evaluated during activities, such as in swimming and gymnastics, can be a potent participation deterrent (Martin Ginis, O'Brien, & Watson, 2003; Wright & Dewar, 1997). Indeed, traditional physical education programs have often been accused of objectifying girls' bodies (Kirk, 1997; Tinning, 1991; Wright, 2000).

Objectification is the experience of being treated as an object to be looked at and evaluated, solely on the basis of physical appearance. According to Fredrickson and Roberts (1997), objectification is so pervasive in our society (e.g., in social and interpersonal encounters where women are more likely to be "checked out" than men, and in the mass media's portrayal of the female body), that women and girls come to think about and value their bodies from an observer's perspective, focusing on what their bodies look like, rather than on unobservable body attributes such as physical abilities and internal sensations. This perspective has been labeled *self-objectification* and is characterized by high levels of self-consciousness and self-monitoring of one's own outward appearance. Furthermore, according to *objectification theory* (Fredrickson & Roberts), self-objectification can account for an array of physical, emotional, and behavioral disturbances that disproportionately affect women, such as depression, eating disorders, and sexual dysfunction.

Objectification is prominent in the two major discourses that underlie traditional physical education (Wright, 2000), specifically, that the body is to be cared for in order to accrue health benefits (which are often equated with physical appearance and thinness) and that the body is to be trained to perform effectively in evaluative, competitive situations. Likewise, physical education has been implicated in the "cult of slenderness," as Tinning (1991) has suggested that objectification and the cultural value ascribed to slim, worked bodies may lead to disordered eating and obsessive exercise regimens. Given these criticisms, there has been a call for examination of alternative physical education practices that teach the pleasure in movement and that strive to integrate the self with the body, rather than practices that objectify and separate the self from the body (e.g., Kirk, 1997; Wright).

With these issues in mind, the purpose of the present study was to examine the effects of an 8-week, girl-friendly physical education and health education module on self-objectification and physique evaluation anxiety (i.e., social physique anxiety) among a sample of Irish girls. The physical education module was designed to provide positive, body-focused experiences that emphasized physical function and kinesthetic movement through creative dance and relaxation exercises, rather than a focus on appearance and sport competition. The health education module was designed to engage students in critical inquiry about the body in contemporary society, building on and extending ongoing research programs focusing on the social construction of the body (e.g., Evans, Davies, & Wright, 2004) and critical inquiry in physical education (e.g., Wright, Macdonald, & Burrows, 2004).

We hypothesized that girls who received the experimental module (experimental condition) would decrease the value placed on appearance-based, objectifying attributes of their bodies and increase the value placed on competence-based, nonobjectifying attributes. No changes in attribute value were expected among the control condition participants. Because objectification theory posits that greater self-objectification is associated with greater appearance anxiety, girls in the experimental condition were also expected to experience a decrease in social physique anxiety, relative to girls in the control condition.

Method

Participants

Four classrooms of students at an all-girls secondary school in Cork City, Ireland, participated in this study—two classes of first-year students ($n = 68$, M age = 12.5) and two classes of Transition Year-1 students ($n = 44$, M age = 15.8). For logistical reasons, participating classes were selected based on the criteria that they had an 80-min physical education class and a 40-min social, personal, and health education (SPHE) class scheduled on the same day. One first-year and one transition-year class were randomly assigned to the experimental condition and the others were assigned to the control condition. (*Transition year* in Irish secondary schools is the year between junior [high school] cycle examinations and senior [high school] cycle and Leaving Certificate examinations.)

Measures

Social physique anxiety was assessed with the revised 9-item Social Physique Anxiety Scale (SPAS; Martin, Rejeski, Leary, McAuley, & Bane, 1997), a questionnaire designed to measure the anxiety that people experience in response to others' evaluations of their bodies (Hart, Leary, & Rejeski, 1989). Participants indicated the extent to which each item was characteristic of them by rating it from 1 (*not at all characteristic of me*) to 5 (*extremely characteristic of me*). Two items were reverse-scored, and then all responses were summed to create a total SPAS score, with higher scores indicating greater social physique anxiety. The SPAS has been previously validated and deemed internally consistent in samples of Irish schoolgirls (Martin, Leary, & O'Brien, 2001; Martin Ginis et al., 2003).

Self-objectification was measured with Noll and Fredrickson's (1998) 10-item Self-Objectification Questionnaire, which assesses the extent to which people regard their bodies in objectifying, appearance-based terms versus nonobjectifying, competence-based terms. The scale consists of five appearance-based physical attributes (physical attractiveness, sex appeal, weight, measurements, and firm sculpted muscles) and five competence-based physical attributes (health, strength, physical coordination, energy level, and physical fitness). Participants ranked the attributes according to personal importance such that the least-valued attribute received a ranking of 0 and the most-valued attribute received a ranking of 9. Changes were examined in rankings of the 10 attributes. Evidence of the scale's construct validity has been provided in a sample of adolescent girls (Slater & Tiggemann, 2002).

Procedure

The University of Limerick Research Ethics Committee approved the study protocol. Each student participant and one of her parents provided informed consent. During the first and last physical education classes of the 8-week study period, the first author administered the questionnaire measures to girls in both study conditions. To ensure participant anonymity, respondents were asked not to write their names on the questionnaires, but rather, to identify themselves using a self-chosen pseudonym.

Classes assigned to the experimental condition then began the 8-week module, taught by the first author (O'Brien), a fully qualified physical education teacher. The physical education component of the module was delivered once per week during an 80-min class and the health education component was delivered during a 40-min class later that day. Classes assigned to the control condition followed their standard physical education and SPHE curriculums delivered by their usual teachers (for curriculum guidelines, see National Council for Curriculum and Assessment, 2000, 2003; Curriculum and Examinations Board, 1993). The amount of time devoted to physical education and SPHE was equivalent across the two conditions.

Overview of the Experimental Module

The purpose of the module was to do "body work" at a practical level and also to cognitively engage students in critical inquiry about the body in contemporary society. The practical work consisted of providing body-focused experiences using relaxation, Feldenkrais method guided imagery exercises, and creative dance. The Feldenkrais method is a form of somatic education that uses gentle movement and directed attention to teach body awareness. Its purpose is to free up the body by abandoning old physical and psychological habits of movement and restrictions on movement, and to widen and refine "the cerebral control of the muscular range" (Feldenkrais, 1985, p. 155). The exercises integrate body and mind, and involve both internal and external parts of the body, as well as body parts that are not usually acknowledged in physical education lessons (e.g., earlobes, underside of the arm; Wright, 2000). The specific activities used were provided by Hutchinson's (1985) therapeutic program, *Transforming Body Image*. Because a heightened self-awareness of outer bodily appearance and a lack of awareness to inner body experience is characteristic of self-objectification (Fredrickson & Roberts, 1997), it was thought that these activities would help the girls become more aware and appreciative of qualities of their bodies that are unrelated to appearance (e.g., physical function, sensation).

Creative dance was used to develop the body-awareness prompted by the Feldenkrais exercises. Following Frank's (1991) work on communicative bodies, creative dance was selected as the most appropriate physical education movement medium through which students might best express body awareness in their own unique ways. Creative dance might also be a form of physical risk taking—a behavior that may help women come to appreciate nonobjectifying qualities of their bodies (Fredrickson & Roberts, 1997).

The critical inquiry component of the module aimed to make girls more aware of the social construction of bodies, how this contributes to self-objectification,

and the consequences of self-objectification. According to objectification theory (Fredrickson & Roberts, 1997), making girls more aware of the adverse effects of objectifying images and treatment can help them to resist objectification, thereby enhancing their psychological well-being. Various activities were used to identify and challenge the assumptions underlying self-objectifying behaviors and ways of thinking, such as workbook exercises, classroom discussions, small group activities, and magazine image analyses. Development of the activities was based on previous work involving adolescent girls (Gorely, Holroyd, & Kirk, 2003; Oliver & Lalik, 2000). Table 1 provides an overview of the topics and activities contained in the experimental module.

Results

Because of the hierarchical nature of the data (students nested in classes), it was necessary to perform a test of nonindependence of the individual observations in order to determine whether the unit of analysis should be at the individual or the classroom level. This was accomplished by conducting a series of multilevel regression models on the pre- and posttest SPAS and Self-Objectification scores in which the random effects of classrooms were tested. The classroom effect was not significant in any of the models (all $ps > .15$). Thus, subsequent analyses utilized the individual as the unit of analysis.

Of the 104 girls registered in the four classes, 10 were absent at pretesting and another 9 were absent at posttesting. The number of girls with complete data is $n = 85$. Pre- and posttest measures are presented in Table 2.

A series of Wilcoxon signed-rank tests were computed to examine whether the experimental and control groups changed their rankings of the competence- and appearance-based attributes. A priori, planned comparison alpha levels were employed for all tests (i.e., $p < .05$). The experimental group significantly increased the importance placed on two competence-based attributes—health and strength ($ps < .05$)—and decreased the value they placed on one appearance-based attribute—sex appeal ($p = .03$). Unexpectedly, the control group decreased the value they placed on weight and physical fitness ($ps < .04$). No other comparisons were significant.

A general linear model repeated-measures ANOVA was conducted to examine changes in SPAS scores. A significant main effect emerged for time, $F(1, 83) = 7.67$, $p < .01$. There was an overall increase in social physique anxiety over the study period; however, post hoc t tests indicated that this increase was significant ($p < .01$) for the control condition only (see Table 2). Neither the main effect for group nor the Group $\times$ Time interaction was significant ($ps > .17$).

Discussion

This experiment examined the effects of a body-focused health and physical education module on girls' self-objectification and social physique anxiety. In partial support of our hypotheses, girls in the experimental condition decreased the priority placed on the objectifying attribute of sex appeal and increased the priority placed on two competence attributes—health and strength. Also, girls in the experimental condition did not experience the significant increases in social physique anxiety

Table 1 Overview of the Pilot Module's Content as Delivered During the Social Personal and Health Education (SPHE) and Physical Education (PE) Classes

Week	PE Class	SPHE Class	SPHE Theory Link ^a
1	• Warm-up and team challenge • Relaxation exercise • Analyze purpose of dance • Identify types of dance	• Activities to promote cooperation • Respect for each contribution • Problem-solving activities for group structure • Critical analysis of interaction • Review of class activities in workbook	Facilitate perceptions of the class as a safe environment for taking physical risks.
2	• Warm-up with childhood games • Imagery exercise, childhood theme • Team challenge • Evaluating “fun” in childhood vs. now • Square dance	• Childhood games identified and students classify motor and social skills learned • Imagery exercise on childhood • Reflective writing in workbooks	Increase awareness of developmental aspects of self-objectification.
3	• Warm-up and team challenge • Relaxation and imagery exercises • Peer teaching of dance steps for swing dance • Full group dance • Contrast square vs Irish dance	• Tasks to promote communication and leadership • Roles and responsibilities in groups • Discussion of task and social orientations • Reflective writing in workbooks	Facilitate perceptions of the class as a safe social environment; develop new ways of interacting and leadership that undermine objectification.
4	• Warm-up and team challenge • Relaxation and imagery session • Small group contemporary dances • Create a dance • Reflect on self-consciousness	• Friendship networks and analysis • Hierarchies and roles girls play • Mapping positive and negative places • Developing a friendship charter • Developing a PE charter	Develop awareness of the body as a central element in girls’ power relations and social landscape; increase awareness of their own objectifying tendencies.

(continued)

Table 1 (continued)

Week	PE Class	SPHE Class	SPHE Theory Link ^a
5	• Warm-up and team challenge	• Menstruation	Increase awareness of health risks associated with self-objectification.
	• Relaxation and imagery exercises	• Attracting the “gaze”	
	• Whole class dance	• The pursuit of perfection	
	• Create a dance to represent “giving and getting the gaze”	• Genetic celebrities and inferior fixes	
6	• Warm-up and team challenge	• Analysis of magazines	Increase awareness of the media’s role in perpetuating the objectification of women and self-objectification.
	• Relaxation and imagery exercises	• Popularity poll	
	• Small group dance; their music choice	• Discussion of masculinity and femininity	
	• Analysis of the movement	• Create and critique collages of magazine photos	
	• Critique of dance		
7	• Warm-up and team challenge	• Identifying self-surveillance techniques	Increase girls’ awareness of their own self-objectifying behaviors.
	• Relaxation and imagery exercises	• Frequency of body-checking behaviors	
	• Replacing surveillance	• Pledge to note and restrict surveillance or checking	
	• Dance and positive imagery	• Discuss in workbook	
8	• Warm-up and team challenge	• Appreciating individual differences	Develop an appreciation of different types of bodies and physical competence rather than appearance.
	• Relaxation and imagery exercises	• Highlights of Special Olympics	
	• Ability to construct/perform	• Celebrating ability not appearance	
	• Final dance	• Communicating not competing bodies	

^aSPHE Theory Link refers to how the SPHE classes were tied to objectification theory.

Table 2 Means, Standard Deviations, and Significance Tests for the Pre- and Posttest Measures of Social Physique Anxiety and Objectified Body Consciousness

Item	Control condition			Experimental condition		
	Pretest	Posttest	<i>p</i>	Pretest	Posttest	<i>p</i>
Social physique anxiety	27.9 ± 7.9	29.9 ± 8.7	<.01	26.1 ± 8.2	27.0 ± 7.5	.32
Rankings of Appearance Attributes						
Weight	6.3 ± 1.9	5.5 ± 2.4	.02	6.0 ± 2.5	6.1 ± 2.4	.87
Sex appeal	2.7 ± 2.6	3.1 ± 3.1	.57	3.7 ± 3.0	2.8 ± 2.4	.03
Physical attractiveness	5.1 ± 2.7	5.2 ± 2.5	.75	5.7 ± 2.6	5.9 ± 2.7	.77
Firm sculpted muscles	1.8 ± 1.9	2.2 ± 2.1	.23	1.8 ± 2.1	1.8 ± 2.4	.77
Measurements	3.1 ± 2.5	3.7 ± 2.7	.09	4.0 ± 2.8	3.2 ± 2.6	.08
Rankings of Competence Attributes						
Health	7.7 ± 2.1	7.8 ± 2.0	.91	6.1 ± 2.4	7.3 ± 2.4	<.01
Strength	3.4 ± 2.3	3.5 ± 2.4	.78	2.5 ± 2.1	3.5 ± 2.3	.04
Energy	4.7 ± 2.3	5.0 ± 2.4	.53	4.8 ± 2.2	4.5 ± 2.2	.30
Fitness	6.0 ± 2.1	5.1 ± 2.4	.04	5.6 ± 2.6	5.6 ± 2.2	.96
Coordination	4.2 ± 2.7	4.0 ± 2.6	.49	4.2 ± 2.8	4.3 ± 2.4	1.00

that occurred within the control group. Together, these findings lend support to objectification theory's (Fredrickson & Roberts, 1997) prediction that through positive experiences of physical movement and physical risk taking, and through education about the causes and effects of objectification, girls can learn to resist self-objectification and its negative psychological consequences.

Although we predicted an increase in priority for *all* of the non-appearance-based attributes (not just health and appearance) and a decrease in priority for *all* of the appearance-based attributes (not just sex appeal), in hindsight, our expectations were probably unrealistic for at least two reasons. First, it is unlikely that an eight-class module would counter the effects of daily competing messages from the media, friends, and family about the importance of appearance. Many studies have shown that people who are more physically attractive derive more favorable social and material outcomes in life than people who are physically unattractive (French, 2002; Langlois et al., 2000). It would be naïve to expect girls to ignore this fact of life. Second, because the self-objectification scale items are ranked relative to one another, in order for girls to increase the priority placed on all competence attributes, they have to simultaneously decrease the priority placed on all appearance attributes. Given the importance of physical attractiveness to life outcomes, it is unlikely that such a shift would occur. However, it may be possible for girls to learn to *equally* value competence and appearance as a result of experiencing the experimental module. Further research is needed to address this question, using measures that can detect such shifts toward parity.

The scaling of the self-objectification measure might also explain the unexpected finding that girls in the control condition showed a significant decrease in

the importance placed on weight. As can be seen in Table 2, at posttest, the control group increased the importance ascribed to all of the appearance-based attributes except weight. Small (nonsignificant) increases in the rankings for the other four appearance attributes could have had a cumulative negative effect on the ranking for weight. Nevertheless, given that most of the objectifying, appearance-based attributes tended to become more important for girls in the control condition, while fitness became significantly less important, it seems that the standard curriculum may actually be doing more harm than good with regards to altering girls' self-objectifying tendencies. This notion is consistent with the position that traditional physical education practices are body objectifying (Kirk et al., 2000; Tinning, 1991; Wright, 2000).

The standard curriculum was also associated with significant increases in social physique anxiety. Puberty has been identified as a time when girls' social physique anxiety levels increase (Thompson & Chad, 2002) possibly in response to rapid changes in their body composition (Eklund & Bianco, 2000) and an increased sensitivity to the evaluation of their appearance by others (Harter, 1999). Consistent with objectification theory (Fredrickson & Roberts, 1997), the experimental module may have prevented developmentally linked increases in social physique anxiety by shifting girls' attention away from their outward appearance to inner, nonappearance body attributes.

Despite these encouraging results, the present investigation has some limitations that warrant mention. First, the sample size was relatively small and limited to girls from a single all-girls school in Ireland. These sample characteristics limit the generalizability of our findings. For example, it is not known whether similar findings would emerge in a coeducational setting (i.e., girl-friendly physical education may be more complicated in coeducational classes). Second, the scope of this study represents a pilot investigation, and, although the results provide valuable information, there are inherent design weaknesses. Different teachers taught the experimental and control classes, which represents a potential study confounder. Additionally, the students in the classes interacted with each other during the module, which violates the assumption of independence of observations. To fully investigate the effects of the module, multiple teachers and classes need to be incorporated into the design so that the class, rather than the student, can be used as the unit of analysis. A third limitation is that the long-term effects of the module on girls' physical activity levels are unknown. Although lower levels of social physique anxiety and self-objectification have been linked with higher levels of physical activity in Irish children adolescents (Jones, Duffy, & Murphy, 1991; Lynch, 1999; Martin et al., 2001; O'Brien & Martin, 1998), the direct impact of this module on physical activity needs to be determined. Nevertheless, our data are encouraging, and they add to international literature on the critical inquiry of girls' bodies in physical education (e.g., Armour, 1999; Gorely et al., 2003).

Importantly, our results suggest that a body-focused physical education and health education module can have positive effects on social physique anxiety and self-objectification. Appropriate content alone, however, cannot guarantee these outcomes. In implementing any curriculum intervention that aims to be girl-friendly, the teacher must assist the students to create safe spaces, including mutual support and respect for the exploration of body-related issues. Further research that qualitatively and quantitatively examines these issues would be of value.

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Jacinta O'Brien was a lecturer and researcher at the University of Limerick's Department of Physical Education and Sport Science from 1986 until her sudden death on February 3, 2004. The data in this manuscript were drawn from her doctoral dissertation, which she was completing under the supervision of David Kirk. Further details regarding the experimental teaching module and study methodology are available from Kirk. We are grateful to Tom Comyns for his assistance with the preparation of this manuscript.

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